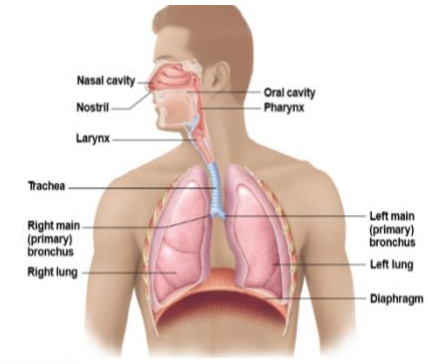
The Respiratory System

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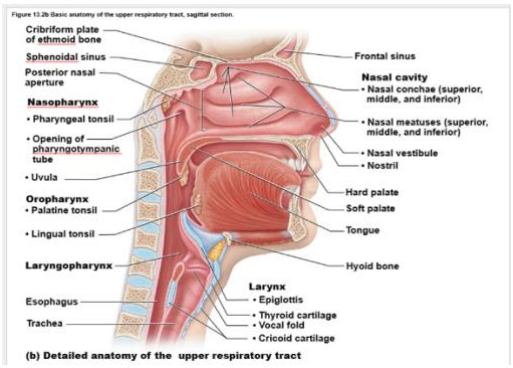
Organs of the Respiratory System

- Nose
- Pharynx
- Larynx
- Trachea
- Bronchi
- Lungs—alveoli



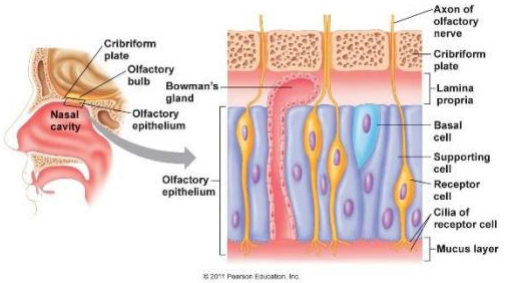
Functional Anatomy of the Respiratory System

- Gas exchanges between the blood and external environment occur only in the alveoli of the lungs
- Upper respiratory tract includes passageways from the nose to larynx
- Lower respiratory tract includes passageways from trachea to alveoli
    - Passageways to the lungs purify, humidify, and warm the incoming air



The Nose

- The only externally visible part of the respiratory system
- Nostrils (nares) are the route through which air enters the nose
- Nasal cavity is the interior of the nose
- Nasal septum divides the nasal cavity
- Olfactory receptors are located in the mucosa on the superior surface
- The rest of the cavity is lined with respiratory mucosa, which
- Moistens air
- Traps incoming foreign particles
- Enzymes in the mucus destroy bacteria chemically
-
- Conchae are projections from the lateral walls
- Increase surface area
- Increase air turbulence within the nasal cavity
    - Increased trapping of inhaled particles
- The palate separates the nasal cavity from the oral cavity
    - Hard palate is anterior and supported by bone
    - Soft palate is posterior and unsupported
- Paranasal sinuses
    - Cavities within the frontal, sphenoid, ethmoid, and maxillary bones surrounding the nasal cavity
    - Sinuses:
        - Lighten the skull
        - Act as resonance chambers for speech
        - Produce mucus

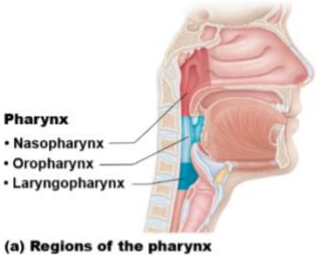


The Pharynx

- Commonly called the *throat*
- Muscular passageway from nasal cavity to larynx
    - Continuous with the posterior nasal aperture
- Three regions of the pharynx
    1. Nasopharynx—superior region behind nasal cavity
    2. Oropharynx—middle region behind mouth
    3. Laryngopharynx—inferior region attached to larynx

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- Oropharynx and laryngopharynx serve as common passageway for air and food
  - Epiglottis routes food into the posterior tube, the esophagus
- Pharyngotympanic tubes open into the nasopharynx
  - Drain the middle ear
- Tonsils are clusters of lymphatic tissue that play a role in protecting the body from infection
  - Pharyngeal tonsil (adenoid), a single tonsil, is located in the nasopharynx
  - Palatine tonsils (2) are located in the oropharynx at the end of the soft palate
  - Lingual tonsils (2) are found at the base of the tongue



(a) Regions of the pharynx

The Larynx

- Commonly called the *voice box*
- Functions
  - Routes air and food into proper channels
  - Plays a role in speech
- Located inferior to the pharynx
- Made of eight rigid hyaline cartilages
  - Thyroid cartilage (Adam's apple) is the largest
- Epiglottis
  - Spoon-shaped flap of elastic cartilage
  - Protects the superior opening of the larynx
  - Routes food to the posteriorly situated esophagus and routes air toward the trachea
  - During swallowing, the epiglottis rises and forms a lid over the opening of the larynx
- Vocal folds (true vocal cords)
  - Vibrate with expelled air
  - Allow us to speak
- The glottis includes the vocal cords and the opening between the vocal cords

The Trachea

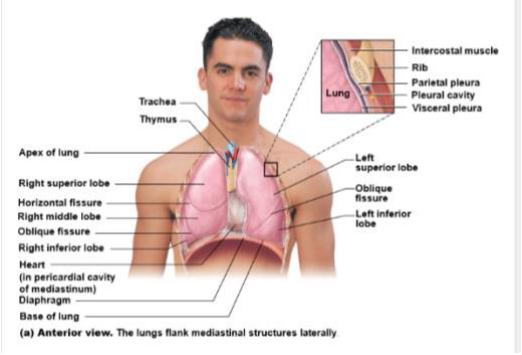
- Commonly called the *windpipe*
- 4-inch-long tube that connects to the larynx
- Walls are reinforced with C-shaped rings of hyaline cartilage, which keep the trachea patent (open)
- Lined with ciliated mucosa
  - Cilia beat continuously in the opposite direction of incoming air
  - Expel mucus loaded with dust and other debris away from lungs

The Main Bronchi

- Formed by division of the trachea
- Each bronchus enters the lung at the hilum (medial depression)
- Right bronchus is wider, shorter, and straighter than left
- Bronchi subdivide into smaller and smaller branches

The Lungs

- Occupy the entire thoracic cavity except for the central *mediastinum*
- Apex of each lung is near the clavicle (superior portion)
- Base rests on the diaphragm
- Each lung is divided into lobes by fissures
  - Left lung—two lobes
  - Right lung—three lobes
- Serosa covers the outer surface of the lungs
  - Pulmonary (visceral) pleura covers the lung surface
  - Parietal pleura lines the walls of the thoracic cavity
- Pleural fluid fills the area between layers
  - Allows the lungs to glide over the thorax
  - Decreases friction during breathing
- Pleural space (between the layers) is more of a potential space



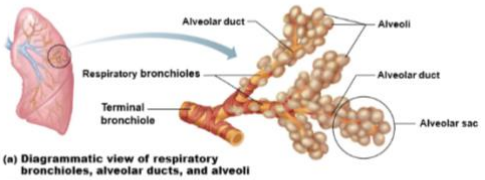
(a) Anterior view. The lungs flank mediastinal structures laterally.

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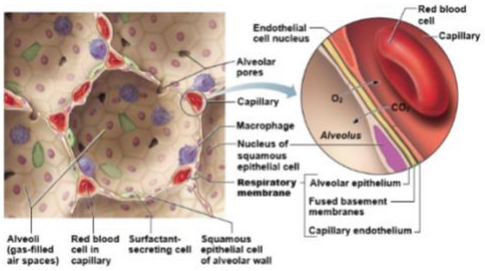
- The bronchial tree
  - Main bronchi subdivide into smaller and smaller branches
  - Bronchial (respiratory) tree is the network of branching passageways
  - All but the smallest passageways have reinforcing cartilage in the walls
  - Conduits to and from the respiratory zone
  - Bronchioles (smallest conducting passageways)

Respiratory Zone Structures and the Respiratory Membrane

- Terminal bronchioles lead into respiratory zone structures and terminate in alveoli
- Respiratory zone includes the:
  - Respiratory bronchioles
  - Alveolar ducts
  - Alveolar sacs
  - Alveoli (air sacs)—the only site of gas exchange
- Conducting zone structures include all other passageways
- Alveoli
  - Simple squamous epithelial cells largely compose the walls
  - Alveolar pores connect neighboring air sacs
- Pulmonary capillaries cover external surfaces of alveoli
- Respiratory membrane (air-blood barrier)
  - On one side of the membrane is air, and on the other side is blood flowing past
  - Formed by alveolar and capillary walls
- Gas crosses the respiratory membrane by diffusion
  - Oxygen enters the blood
  - Carbon dioxide enters the alveoli
- Alveolar macrophages ("dust cells")
  - Add protection by picking up bacteria, carbon particles, and other debris
- Surfactant (a lipid molecule)
  - Coats gas-exposed alveolar surfaces
  - Secreted by cuboidal surfactant-secreting cells



(a) Diagrammatic view of respiratory bronchioles, alveolar ducts, and alveoli

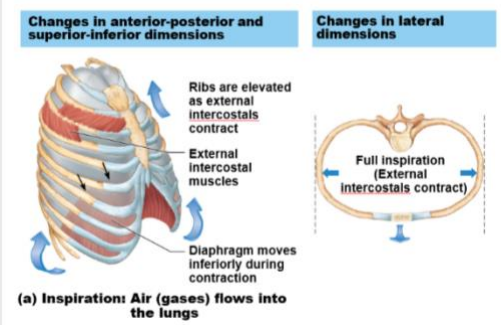


Respiratory Physiology

- Functions of the respiratory system
  - Supply the body with oxygen
  - Dispose of carbon dioxide
- Four events of respiration
  1. Pulmonary ventilation—moving air into and out of the lungs (commonly called *breathing*)
  2. External respiration—gas exchange between pulmonary blood and alveoli
     - Oxygen is loaded into the blood
     - Carbon dioxide is unloaded from the blood
  3. Respiratory gas transport—transport of oxygen and carbon dioxide via the bloodstream
  4. Internal respiration—gas exchange between blood and tissue cells in systemic capillaries

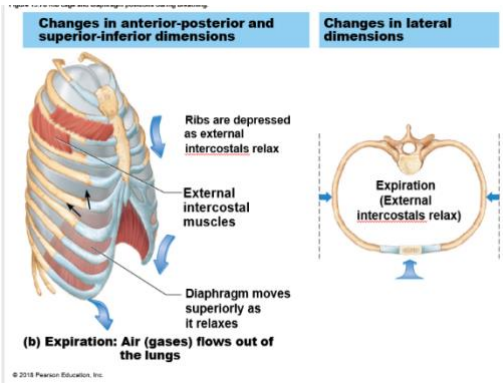
Mechanics of Breathing

- Pulmonary ventilation
  - Mechanical process that depends on volume changes in the thoracic cavity
  - Volume changes lead to pressure changes, which lead to the flow of gases to equalize pressure
- Two phases of pulmonary ventilation
  - Inspiration = inhalation
    - Flow of air into lungs
  - Expiration = exhalation
    - Air leaving lungs
    - Inspiration (inhalation)
  - Diaphragm and external intercostal muscles contract



(a) Inspiration: Air (gases) flows into the lungs

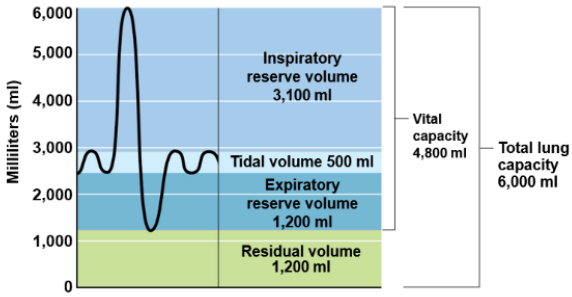
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- Intrapulmonary volume *increases*
- Gas pressure *decreases*
- Air flows into the lungs until intrapulmonary pressure equals atmospheric pressure
  - Expiration (exhalation)
- Largely a passive process that depends on natural lung elasticity
- Intrapulmonary volume *decreases*
- Gas pressure *increases*
- Gases passively flow out to equalize the pressure
- Forced expiration can occur mostly by contraction of internal intercostal muscles to depress the rib cage

- Intrapleural pressure
  - The pressure within the pleural space) is *always* negative
  - Major factor preventing lung collapse
  - If intrapleural pressure equals atmospheric pressure, the lungs recoil and collapse

Respiratory Volumes and Capacities



- Factors affecting respiratory capacity
  - Size
  - Sex
  - Age
  - Physical condition
- Tidal volume (TV)
  - Normal quiet breathing
  - 500 ml of air is moved in/out of lungs with each breath
- Inspiratory reserve volume (IRV)
  - Amount of air that can be taken in forcibly over the tidal volume
  - Usually around 3,100 ml
- Expiratory reserve volume (ERV)
  - Amount of air that can be forcibly exhaled after a tidal expiration
  - Approximately 1,200 ml
- Residual volume
  - Air remaining in lung after expirationCannot be voluntarily exhaled
  -
  - Allows gas exchange to go on continuously, even between breaths, and helps keep alveoli open (inflated)
  - About 1,200 ml
- Vital capacity
  - The total amount of exchangeable air
  - Vital capacity = TV + IRV + ERV
  - 4,800 ml in men; 3,100 ml in women
- Dead space volume
  - Air that remains in conducting zone and never reaches alveoli
  - About 150 ml
- Functional volume
  - Air that actually reaches the respiratory zone
  - Usually about 350 ml
- Respiratory capacities are measured with a spirometer

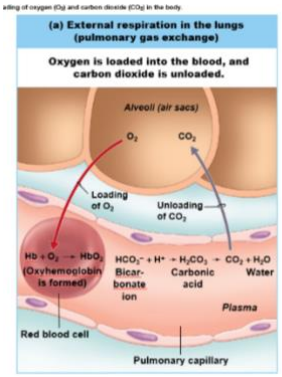
Nonrespiratory Air Movements

- Can be caused by reflexes or voluntary actions
- Examples
  - Cough and sneeze—clears lungs of debris
  - Crying—emotionally induced mechanism
  - Laughing—similar to crying
  - Hiccup—sudden inspirations
  - Yawn—very deep inspiration

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External Respiration, Gas Transport, and Internal Respiration

- Gas exchanges occur as a result of diffusion
  - External respiration is an exchange of gases occurring between the alveoli and pulmonary blood (pulmonary gas exchange)
  - Internal respiration is an exchange of gases occurring between the blood and tissue cells (systemic capillary gas exchange)
- Movement of the gas is toward the area of lower concentration

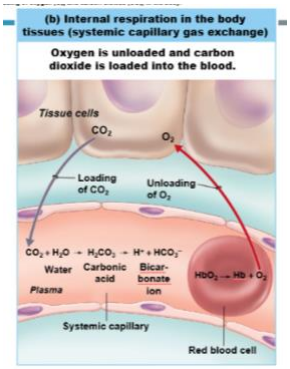


External Respiration

- Oxygen is loaded into the blood
  - Oxygen diffuses from the oxygen-rich air of the alveoli to the oxygen-poor blood of the pulmonary capillaries
- Carbon dioxide is unloaded out of the blood
  - Carbon dioxide diffuses from the blood of the pulmonary capillaries to the alveoli

Gas Transport in the Blood

- Oxygen transport in the blood
  - Most oxygen travels attached to hemoglobin and forms oxyhemoglobin (HbO_2)
  - A small dissolved amount is carried in the plasma
- Carbon dioxide transport in the blood
  - Most carbon dioxide is transported in the plasma as bicarbonate ion (HCO_3^-)
  - A small amount is carried inside red blood cells on hemoglobin, but at different binding sites from those of oxygen
- For carbon dioxide to diffuse out of blood into the alveoli, it must be released from its bicarbonate form:
  - Bicarbonate ions enter RBC
  - Combine with hydrogen ions
  - Form carbonic acid (H_2CO_3)
  - Carbonic acid splits to form water + CO_2
  - Carbon dioxide diffuses from blood into alveoli

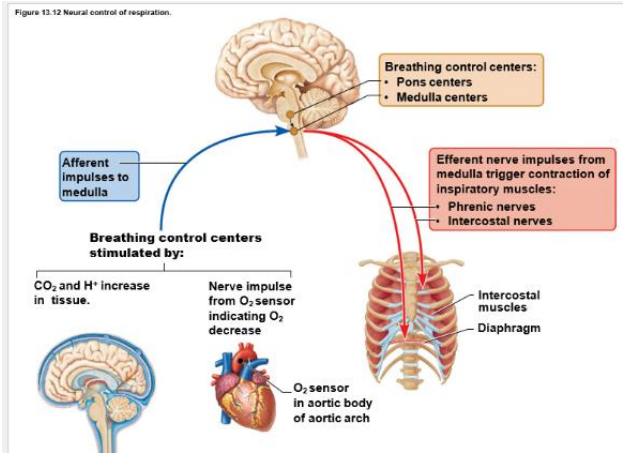


Internal Respiration

- Exchange of gases between blood and tissue cells
- An opposite reaction from what occurs in the lungs
  - Carbon dioxide diffuses out of tissue cells to blood (called *loading*)
  - Oxygen diffuses from blood into tissue (called *unloading*)

Control of Respiration

- Neural regulation: setting the basic rhythm
  - Activity of respiratory muscles is transmitted to and from the brain by phrenic and intercostal nerves
  - Neural centers that control rate and depth are located in the medulla and pons
    - Medulla—sets basic rhythm of breathing and contains a pacemaker (self-exciting inspiratory center) called the *ventral respiratory group (VRG)*
    - Pons—smoothes out respiratory rate



Control of Respiration

- Normal respiratory rate (eupnea)
  - 12 to 15 respirations per minute
- Hyperpnea
  - Increased respiratory rate, often due to extra oxygen needs

Control of Respiration

- Non-neural factors influencing respiratory rate and depth
  - Physical factors
    - Increased body temperature
    - Exercise
    - Talking

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        - Coughing
    - Volition (conscious control)
    - Emotional factors such as fear, anger, and excitement
    - Chemical factors: CO_2 levels
        - The body's need to rid itself of CO_2 is the *most* important stimulus for breathing
        - Increased levels of carbon dioxide (and thus, a decreased or acidic pH) in the blood increase the rate and depth of breathing
        - Changes in carbon dioxide act directly on the medulla oblongata
    - Chemical factors: oxygen levels
        - Changes in oxygen concentration in the blood are detected by chemoreceptors in the aorta and common carotid artery
        - Information is sent to the medulla
        - Oxygen is the stimulus for those whose systems have become accustomed to high levels of carbon dioxide as a result of disease
        - Hyperventilation
            - Rising levels of CO_2 in the blood (acidosis) result in faster, deeper breathing
            - Exhale more CO_2 to elevate blood pH
            - May result in apnea and dizziness and lead to alkalosis
        - Hypoventilation
            - Results when blood becomes alkaline (alkalosis)
            - Extremely slow or shallow breathing
            - Allows CO_2 to accumulate in the blood

Respiratory Disorders

- Chronic obstructive pulmonary disease (COPD)
    - Exemplified by chronic bronchitis and emphysema
    - Shared features of these diseases
    1. Patients almost always have a history of smoking
    2. Labored breathing (dyspnea) becomes progressively worse
    3. Coughing and frequent pulmonary infections are common
    4. Most COPD patients are hypoxic, retain carbon dioxide and have respiratory acidosis, and ultimately develop respiratory failure
- Chronic bronchitis
    - Mucosa of the lower respiratory passages becomes severely inflamed
    - Excessive mucus production impairs ventilation and gas exchange
    - Patients become cyanotic and are sometimes called "blue bloaters" as a result of chronic hypoxia and carbon dioxide retention
- Emphysema
    - Alveoli walls are destroyed; remaining alveoli enlarge
    - Chronic inflammation promotes lung fibrosis, and lungs lose elasticity
    - Patients use a large amount of energy to exhale; some air remains in the lungs
    - Sufferers are often called "pink puffers" because oxygen exchange is efficient
    - Overinflation of the lungs leads to a permanently expanded barrel chest
    - Cyanosis appears late in the disease
- Lung cancer
    - Leading cause of cancer death for men and women
    - Nearly 90 percent of cases result from smoking
    - Aggressive cancer that metastasizes rapidly
    - Three common types
        1. Adenocarcinoma
        2. Squamous cell carcinoma
        3. Small cell carcinoma
- Asthma
    - Chronically inflamed, hypersensitive bronchiole passages
    - Respond to irritants with dyspnea, coughing, and wheezing

Developmental Aspects of the Respiratory System

- Lungs do not fully inflate until 2 weeks after birth

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    - This change from nonfunctional to functional respiration depends on surfactant
    - Surfactant lowers surface tension so the alveoli do not collapse
    - Surfactant is formed late in pregnancy, around 28 to 30 weeks
- Respiratory rate changes throughout life
    - Newborns: 40 to 80 respirations per minute
    - Infants: 30 respirations per minute
    - Age 5: 25 respirations per minute
    - Adults: 12 to 18 respirations per minute
    - Rate often increases again in old age
- Youth and middle age
    - Most respiratory system problems are a result of external factors, such as infections and substances that physically block respiratory passageways
- Aging effects
    - Elasticity of lungs decreases
    - Vital capacity decreases
    - Blood oxygen levels decrease
    - Stimulating effects of carbon dioxide decrease
    - Elderly are often hypoxic and exhibit sleep apnea
    - More risks of respiratory tract infection

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